

INDU 211: Introduction to Production & Manufacturing Systems

Problem Solution Guide 07: Statistical Quality Control & Process Capability

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[!NOTE]

- **YouTube Video Tutorial:** [Watch Video 7: INDU 211 - Quality Control Problem](#)
- **Video ID:** 1BcAZosLMb0 · **Duration:** 18:10
- **Target Exam Scope:** ❏ **FINAL EXAM (Weeks 7–12, Chapters 14, 15, 8, 17)**
- **Curriculum Context:** Chapter 8 — Quality Control & Statistical Process Control (SPC)

1. Problem Description & Initial Process Data

Bestwood Manufacturing produces short wooden beams with a nominal design length of 24.0 cm. Quality engineers sampled 15 subgroups ($m = 15$), with each subgroup containing 4 individual beams ($n = 4$):

- **Initial Process Summary Statistics:**
 - Number of subgroups: $m = 15$
 - Subgroup sample size: $n = 4$
 - Initial grand mean (average of subgroup averages): $\bar{\bar{X}}_{\text{initial}} = 24.00$ cm
 - Initial average range: $\bar{R}_{\text{initial}} = 1.413$ cm
 - **Control Chart Out-of-Control Identification:**
 - Initial $\bar{X}$ and R charts revealed that **Sample 4** ($\bar{X}_{\{4\}} = 26.15 \text{ cm}$) and **Sample 10** ($\bar{X}_{\{10\}} = 22.75 \text{ cm}$) exceeded the 3-sigma control limits.
 - Assignable causes were identified and corrected. Standard industrial quality protocol requires eliminating these two out-of-control subgroups and recalculating revised in-control process parameters using the remaining $m = 13$ subgroups.
 - **Engineering Design Tolerances:**
 - Nominal length: $24.0 \text{ cm} \pm 2.2 \text{ cm}$
 - Upper Specification Limit: $\text{USL} = 24.0 + 2.2 = 26.2 \text{ cm}$
 - Lower Specification Limit: $\text{LSL} = 24.0 - 2.2 = 21.8 \text{ cm}$
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2. Step-by-Step Worked Solutions

Part 4.1: Revised In-Control Average Length ($\bar{\bar{X}}_{\text{revised}}$)

- **Rapid Calculation Technique:**

1. Calculate initial sum of all 15 subgroup means: $\sum_{i=1}^{15} \bar{X}_i$
 $i = m \times \bar{X}$

$$\{\text{initial}\} = 15 \times 24.00 = 360.00 \text{ cm}$$

2. Subtract the two out-of-control sample means: $\sum_{\text{revised}} \bar{X}_i = 360.00 - \bar{X}_4 - \bar{X}_{10}$

$$\{10\} = 360.00 - 26.15 - 22.75 = 311.10 \text{ cm}$$

3. Divide by the revised number of in-control subgroups ($m_{\text{new}} = 15 - 2 = 13$):

$$\bar{\bar{X}}_{\text{revised}} = \frac{311.10}{13} = \mathbf{23.93 \text{ cm}}$$

- **Conclusion:** When the process is operating in a state of statistical control, the true average beam length produced is 23.93 cm.
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Part 4.2: Estimation of Process Standard Deviation ($\hat{\sigma}$)

- **Theoretical SPC Formula:** The population standard deviation is unbiasedly estimated from the average range using the Hartley control chart factor d_2 :

$$\hat{\sigma} = \frac{\bar{R}}{d_2}$$

- **Hartley Constant Lookup:** From the Standard Control Chart Constants table for subgroup size $n = 4$:

$$d_2 = 2.059$$

- **Calculation:**

$$\hat{\sigma} = \frac{1.413}{2.059} = \mathbf{0.686 \text{ cm}}$$

Part 4.3: Process Capability Ratio (C_p) & Industrial Interpretation

- **Process Capability Ratio Formula:**

$$C_p = \frac{USL - LSL}{6\hat{\sigma}}$$

- **Calculation:**

$$C_p = \frac{26.2 - 21.8}{6 \times 0.686} = \frac{4.40}{4.116} = \mathbf{1.07}$$

- **Industrial Engineering Interpretation:**

- **The 1.33 Standard:** In industrial manufacturing, an existing production process requires a minimum $C_p \geq 1.33$ (a 4-sigma safety margin) to be certified as capable of producing within specifications.
- **Assessment:** Because $C_p = 1.07 < 1.33$, the beam manufacturing process is **NOT CAPABLE** of consistently satisfying the ± 2.2 cm tolerance requirements.
- **Remedial Action:** Bestwood quality engineering must either reduce process variability (σ) by recalibrating cutting equipment or negotiate wider design tolerances with the client.